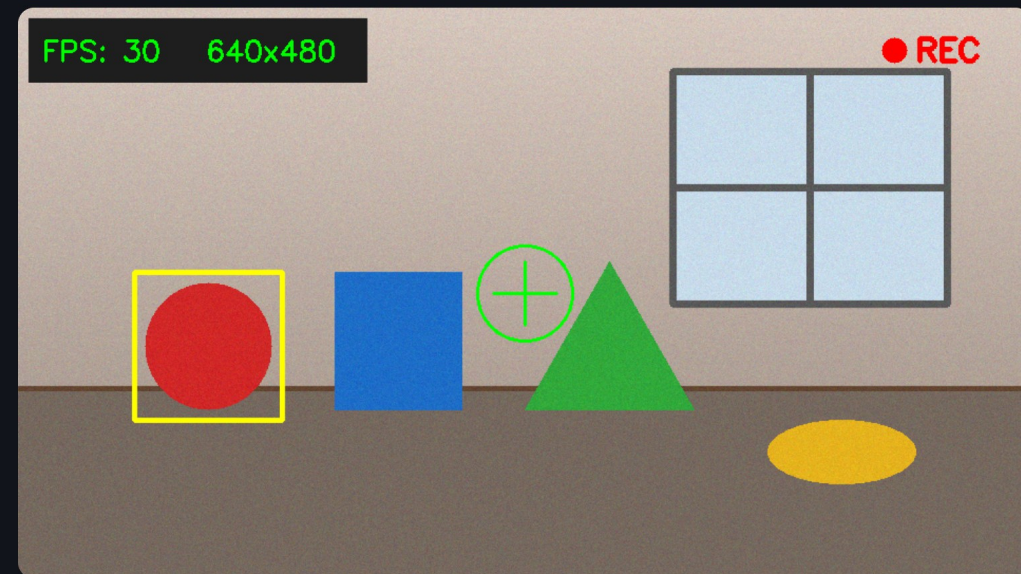


Amaliy dars · 2 soat

OpenCV bilan kamera ustida ishlash

Kameradan kadr olishdan harakat va yuzni
aniqlashgacha — Python va OpenCV yordamida

[Ma'ruzachi ismi] · [Sana]



Dars rejasi — 120 daqiqa

#	Bo'lim	Vaqt
1	Kirish: kompyuter ko'rishi va OpenCV	10 daq
2	Kamerani ishga tushiramiz	15 daq
3	Kadrni o'zgartirish, chizish, yozib olish	15 daq
—	Tanaffus	5 daq
4	Filtrlar: blur, Canny, threshold, effektlar	20 daq
5	Rang bo'yicha aniqlash va shakllar	15 daq
6	Harakatni aniqlash	15 daq
7	Yuzni aniqlash	10 daq
8	Amaliyot, xatolar, savollar	15 daq

Kompyuter ko'rish nima?

Kompyuterga tasvirni «ko'rish» va undan **ma'no chiqarish**ni o'rgatish



Avtomobillar

Yo'l belgilari, chiziqlar va
piyodalarni taniydi



Telefonlar

Yuz orqali qulfni ochadi, rasmni
avtomatik tahrirlaydi



Ishlab chiqarish

Konveyerdagi nuqsonli
mahsulotni topadi

OpenCV — ko'rish uchun tayyor asboblarni qutisi

Open Source Computer Vision Library — tasvir va video bilan ishlash uchun bepul, ochiq kodli kutubxona.

1999

Intel kompaniyasida boshlangan loyiha

2500+

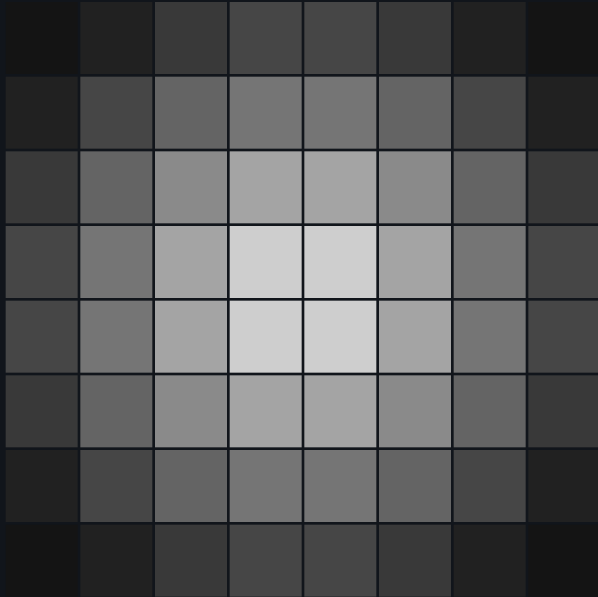
Tayyor, optimallashtirilgan algoritmlar

Python

C++, Java va boshqa tillarda ham ishlaydi

Bugun Python orqali ishlaymiz: `import cv2` — va hamma narsa tayyor.

Rasm — bu raqamlar jadvali



Kulrang rasm: har bir piksel — bitta son, 0 (qora) dan 255 (oq) gacha.

Rangli rasm: har piksel — 3 ta son. OpenCV'da tartib **B, G, R** (RGB emas!).

```
>>> frame.shape
(480, 640, 3)    # balandlik, kenglik, kanal
>>> frame[100, 200]
[ 35  42 210]    # B, G, R — qizil piksel
```

01

Kamerani ishga tushiramiz

O'rnatish · birinchi dastur · video sikli · sozlamalar ⌚ 15 daqiqa

O'rnatish — bitta buyruq

```
# Terminal (cmd) da:  
pip install opencv-python numpy  
  
# Tekshirish:  
python -c "import cv2; print(cv2.__version__)"
```



Kerak bo'ladi

Python 3, veb-kamera (noutbukniki ham bo'ladi), VS Code yoki PyCharm



Qo'shimcha

opencv-contrib-python — qo'shimcha modullar bilan to'liq versiya

Birinchi dastur: jonli video

```
import cv2

cap = cv2.VideoCapture(0)    # 0 — birinchi kamera

while True:
    ok, frame = cap.read()   # bitta kadr
    if not ok:
        break
    cv2.imshow("Kamera", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break                # q — chiqish

cap.release()                # kamerani bo'shatish
cv2.destroyAllWindows()
```



VideoCapture(0)

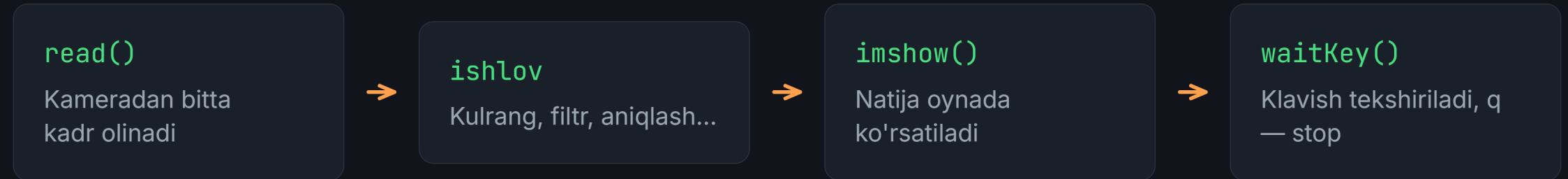
Kamerani ochadi. Ikkinchi kamera — 1, video fayl — "video.mp4"



waitKey(1)

1 ms klavishni kutadi. Usiz oyna qotib qoladi!

Video — tez almashadigan rasmlar



~3
0

marta soniyada shu sikl takrorlanadi. Ishlov qancha og'ir bo'lsa, FPS shuncha tushadi — shuning uchun kadrni kichraytirish foydali.

Kamera sozlamalari

Xususiyat	Nima qiladi
<code>CAP_PROP_FRAME_WIDTH</code>	Kadr kengligi (masalan 1280)
<code>CAP_PROP_FRAME_HEIGHT</code>	Kadr balandligi (masalan 720)
<code>CAP_PROP_FPS</code>	Soniyadagi kadrlar soni
<code>CAP_PROP_BRIGHTNESS</code>	Yorqinlik
<code>cap.isOpened()</code>	Kamera ochildimi? True / False

```
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)

w = cap.get(cv2.CAP_PROP_FRAME_WIDTH)
print("Kenglik:", w)
```

Eslatma: har bir kamera hamma sozlamani qo'llab-quvvatlamaydi — `get()` bilan tekshiring.

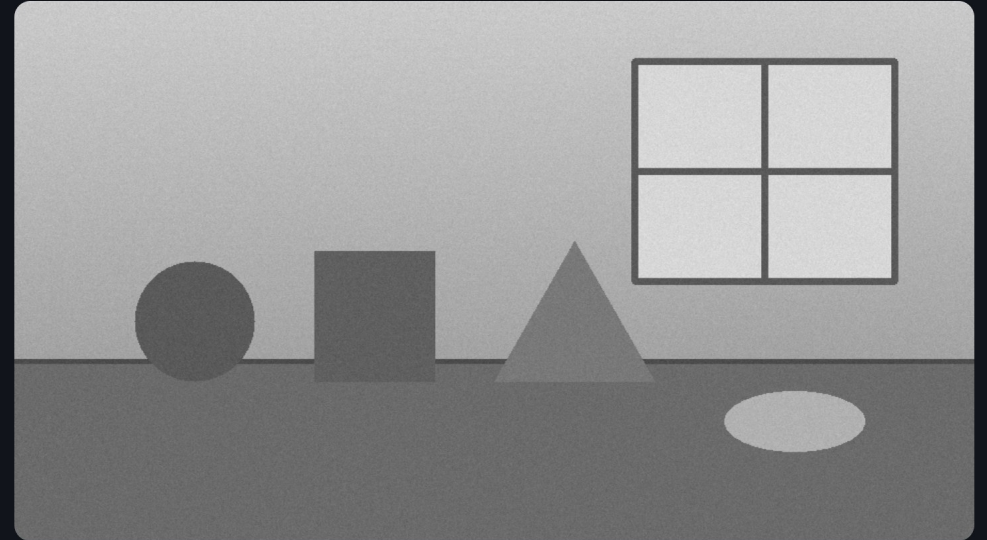
02

Kadrni o'zgartiramiz

Aylantirish · kulrang · chizish · videoni yozib olish ⌚ 15 daqiqa

Aylantirish, o'lcham, kulrang

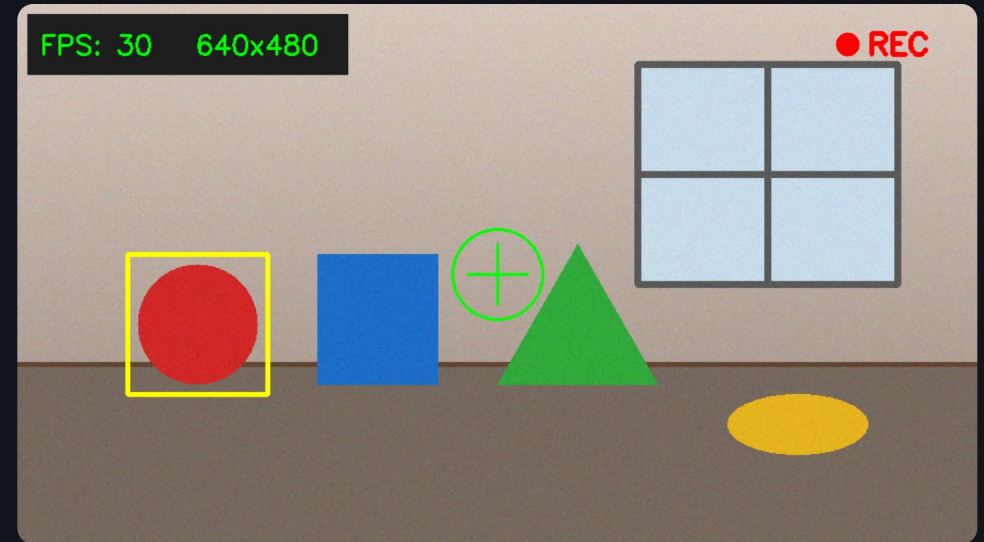
```
# while sikli ichida:  
frame = cv2.flip(frame, 1)          # ko'zgu effekti  
  
small = cv2.resize(frame, (640, 360))  
  
gray = cv2.cvtColor(frame,  
                        cv2.COLOR_BGR2GRAY)  
cv2.imshow("Kulrang", gray)
```



Natija: 3 kanal → 1 kanal. Ma'lumot 3 baravar kam!

Kadrga chizish: matn, ramka, nishon

```
cv2.rectangle(frame, (110, 250), (250, 390),  
               (0, 255, 255), 3) # sariq ramka  
cv2.circle(frame, (480, 270), 45,  
            (0, 255, 0), 2)      # yashil doira  
cv2.line(frame, (450, 270), (510, 270),  
          (0, 255, 0), 2)  
cv2.putText(frame, "FPS: 30", (22, 50),  
            cv2.FONT_HERSHEY_SIMPLEX,  
            0.9, (0, 255, 0), 2)
```



Natija: OpenCV chizgan interfeys elementlari

Videoni yozib olish va skrinshot

```
fourcc = cv2.VideoWriter_fourcc(*"mp4v")
out = cv2.VideoWriter("video.mp4", fourcc,
                      20.0, (640, 480))

n = 0
while True:
    ok, frame = cap.read()
    frame = cv2.resize(frame, (640, 480))
    out.write(frame)                # yozish
    cv2.imshow("REC", frame)
    key = cv2.waitKey(1) & 0xFF
    if key == ord('s'):              # skrinshot
        cv2.imwrite(f"rasm_{n}.png", frame); n += 1
    if key == ord('q'):
        break
out.release()
```

q

Dasturdan chiqish

s

Kadrni PNG qilib saqlash

Muhim

Kadr o'lchami VideoWriter'dagi
bilan bir xil bo'lsin



Tanaffus

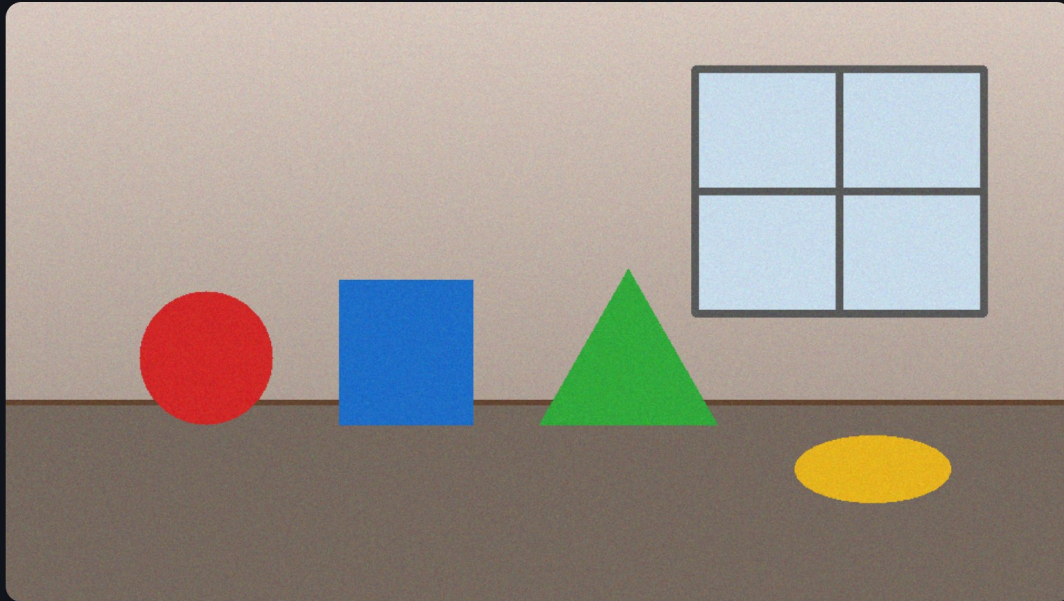
5 daqiqa · Kameraniz ishlayotganini tekshirib qo'ying

03

Filtrlar: kadrni boshqacha ko'ramiz

Blur · Canny chegaralari · threshold · qiziqarli effektlar 🕒 20 daqiqa

Blur: shovqinni yumshatish



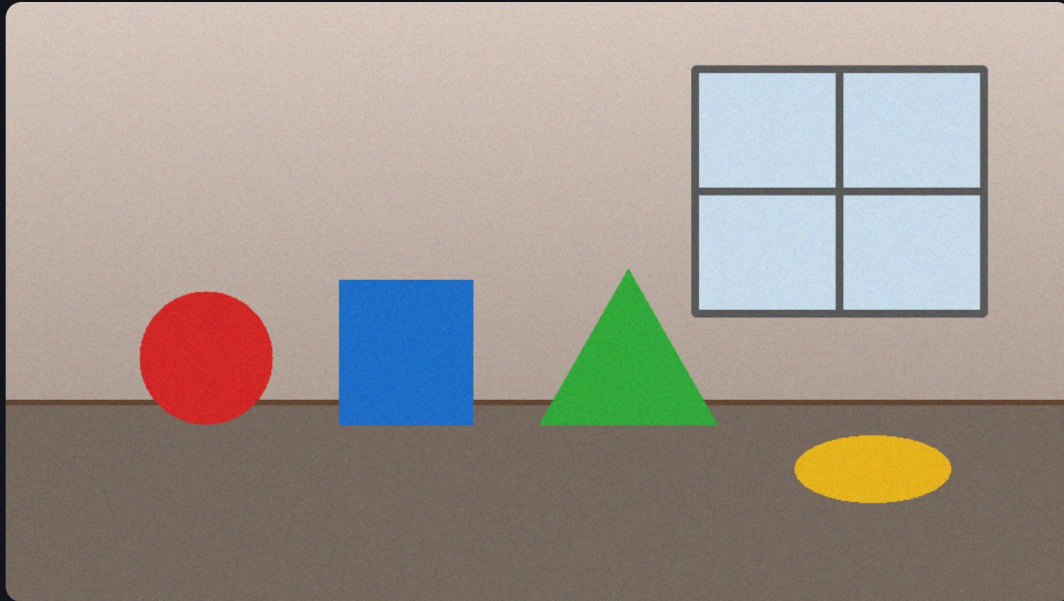
Asl kadr (shovqinli)



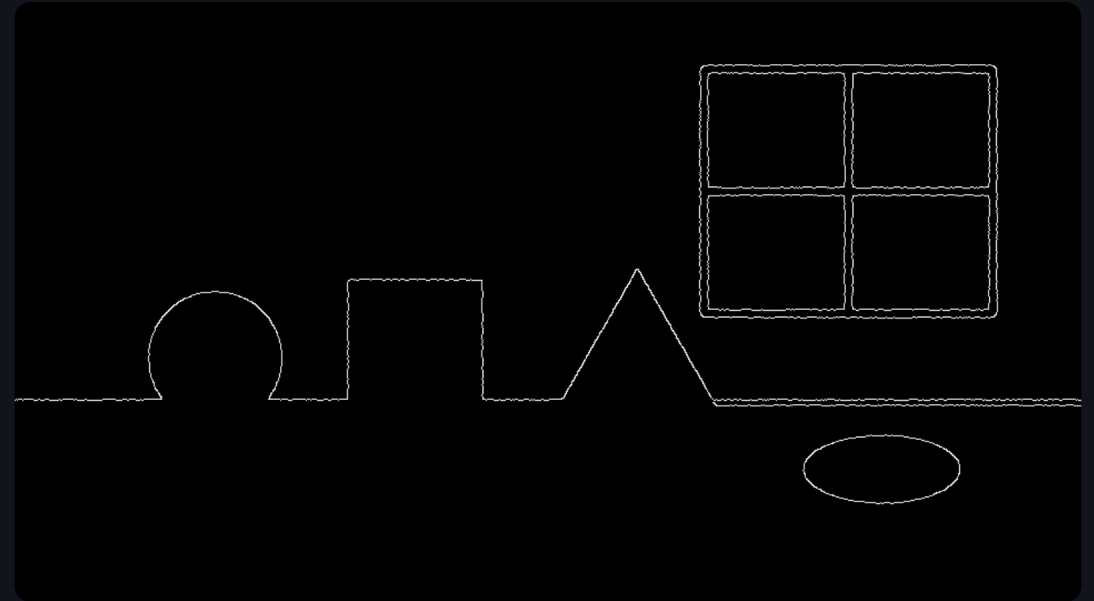
GaussianBlur natijasi

```
blur = cv2.GaussianBlur(frame, (21, 21), 0)
```

Canny: narsalarning chegaralari



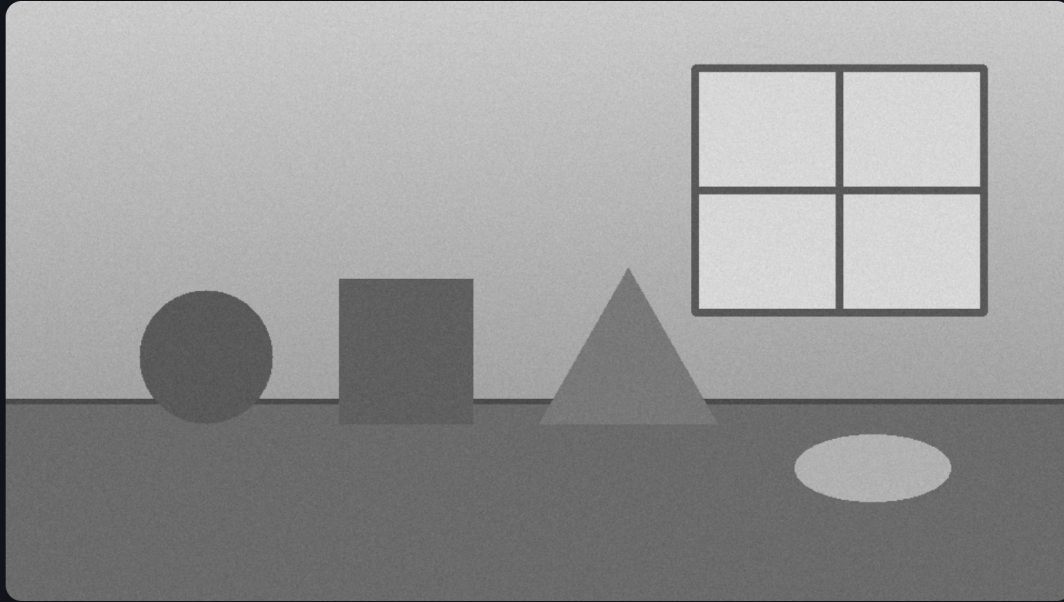
Asl kadr



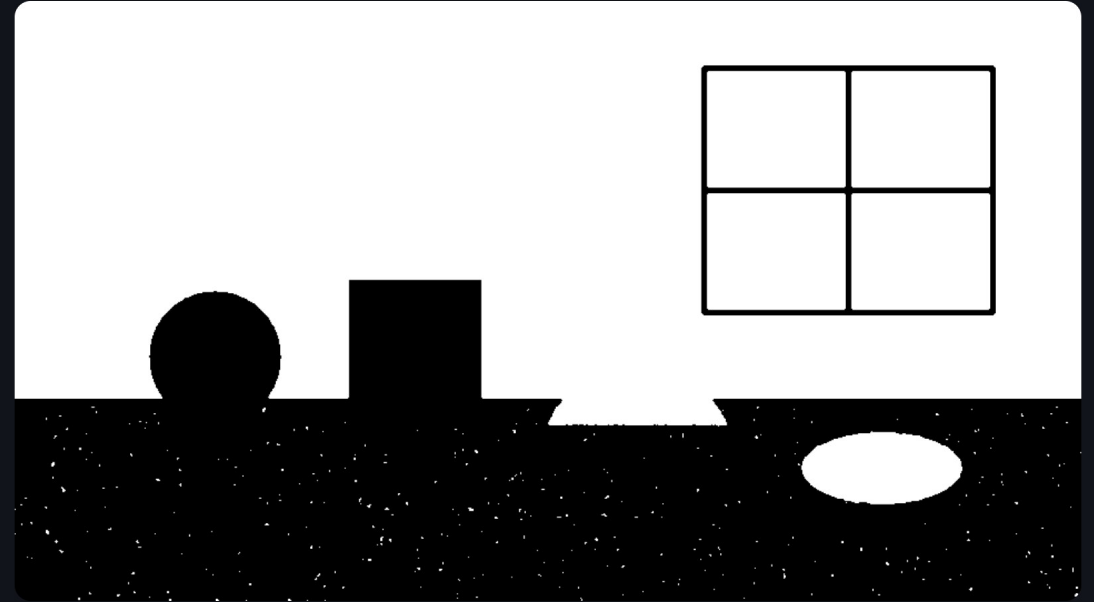
Canny natijasi — faqat chegaralar

```
edges = cv2.Canny(gray, 50, 150)
```

Threshold: oq yoki qora



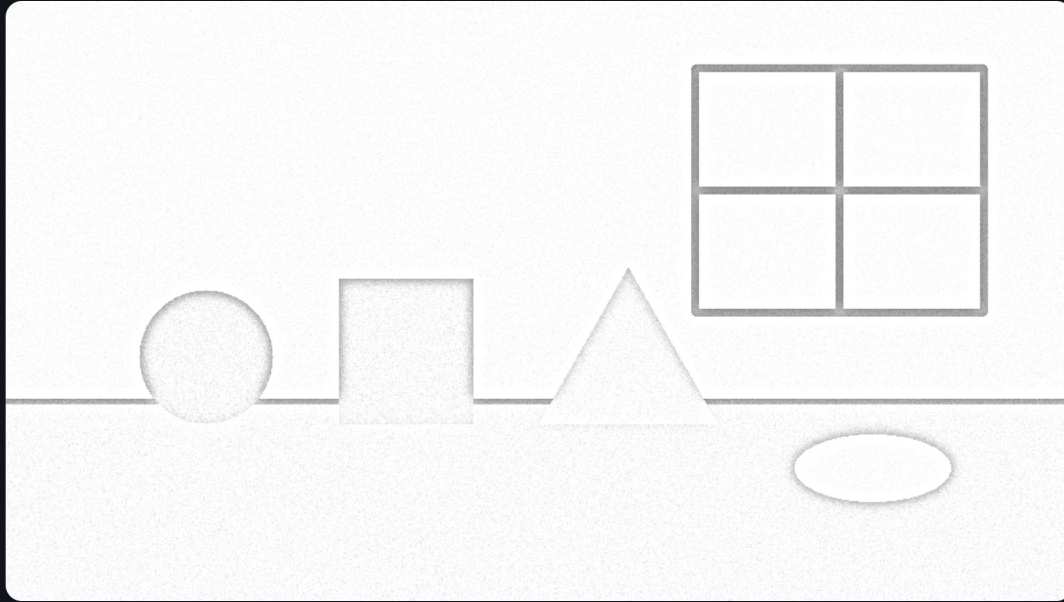
Kulrang kadr



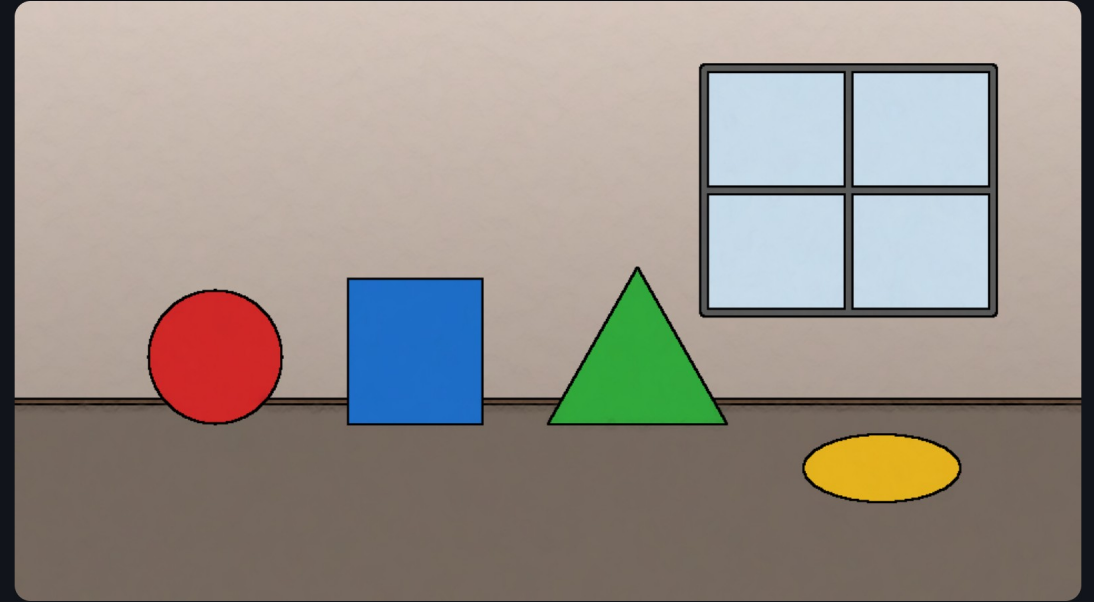
110 dan yorug' → oq, qolgani → qora

```
_, mask = cv2.threshold(gray, 110, 255, cv2.THRESH_BINARY)
```

Qiziqarli effektlar: qalamchizma va multfilm



Qalamchizma



Multfilm (stylization)

```
sketch = cv2.divide(gray, 255 - blur_inv, scale=256)
```

04

Aqli aniqlash

Rang bo'yicha · shakllar · harakat · yuz 🕒 40 daqiqa

Nega HSV? Rangni barqaror tanish uchun



0 qizil

30 sariq

60 yashil

90 moviy

120 ko'k

150 pushti

179



H — ton

Qaysi rang: 0...179. Yorug'lik o'zgarsa ham deyarli o'zgarmaydi



S — to'yinganlik

Rang qanchalik «toza»: 0 kulrang, 255 yorqin



V — yorqinlik

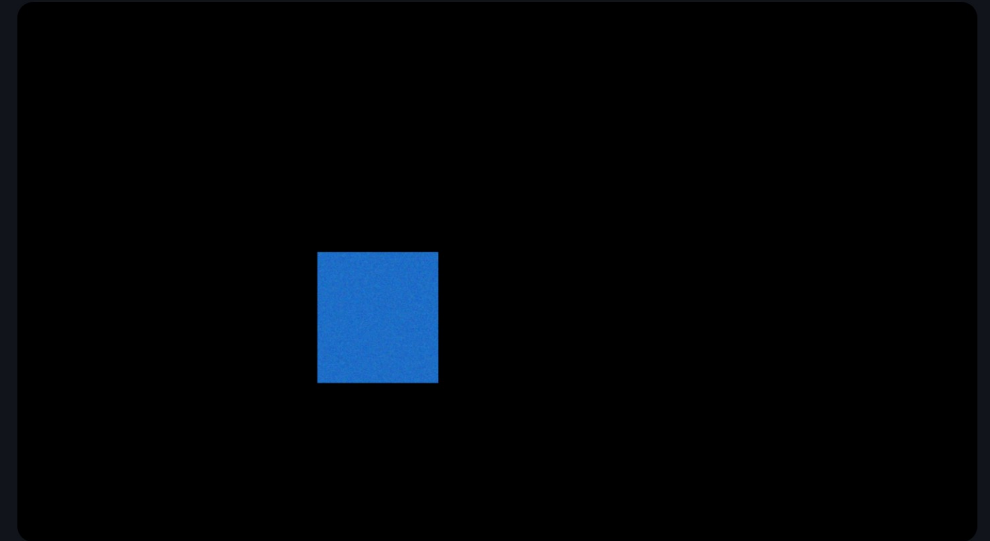
Qanchalik yorug': 0 qora, 255 yorqin

Ko'k narsani topamiz

```
hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

lower = (100, 120, 80)    # ko'k: past chegara
upper = (130, 255, 255)   # ko'k: yuqori chegara
mask = cv2.inRange(hsv, lower, upper)

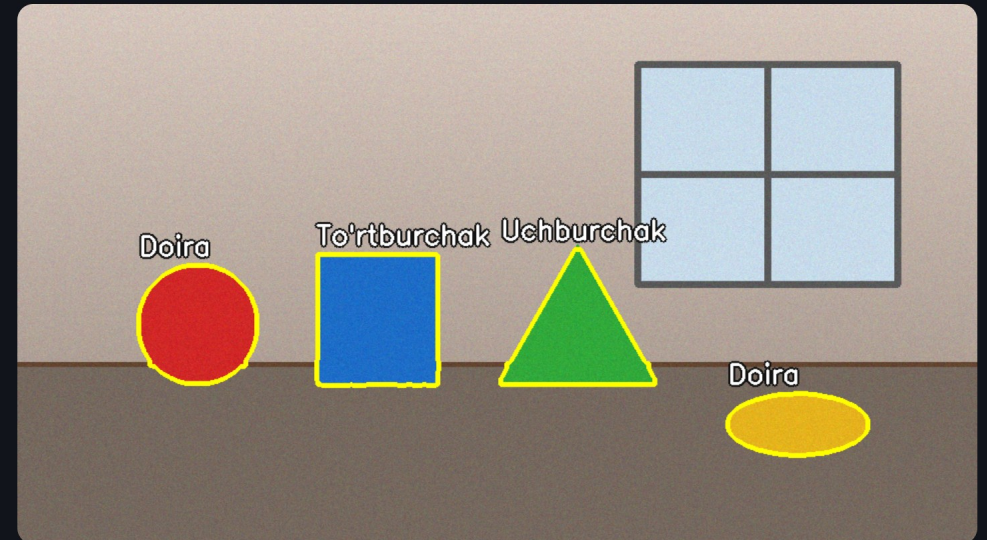
result = cv2.bitwise_and(frame, frame,
                          mask=mask)
```



Natija: kadrдан faqat ko'k quti qoldi

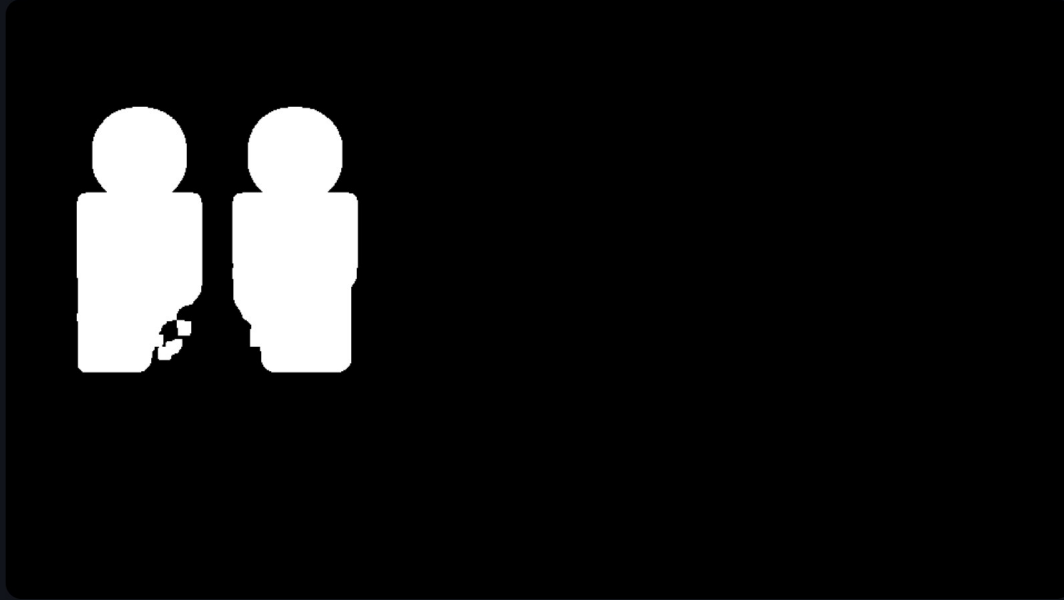
Konturlar: shaklni aniqlash

```
cnts, _ = cv2.findContours(mask,  
    cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)  
  
for c in cnts:  
    if cv2.contourArea(c) < 1500:  
        continue                # mayda shovqin  
    p = cv2.arcLength(c, True)  
    ap = cv2.approxPolyDP(c, 0.03 * p, True)  
    n = len(ap)                  # burchaklar soni  
    name = {3: "Uchburchak",  
            4: "To'rtburchak"}.get(n, "Doira")  
    x, y, w, h = cv2.boundingRect(c)  
    cv2.putText(frame, name, (x, y - 10), ...)
```

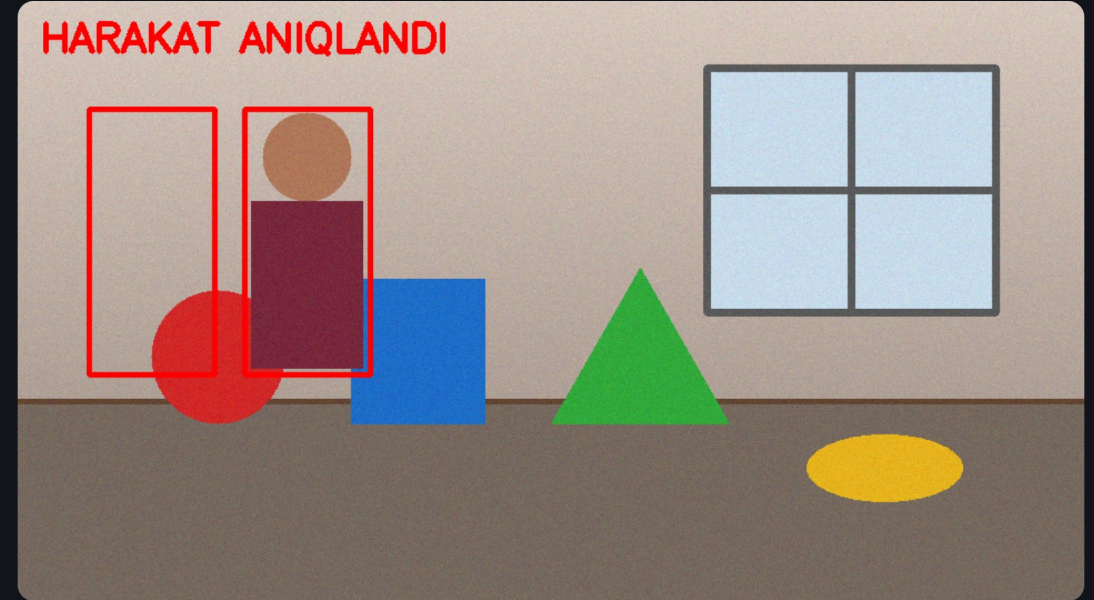


Natija: har bir narsa nomi bilan belgilandi

Harakatni aniqlash: 3 qadam



1–2. Ikki kadr farqi (absdiff) → threshold



3. Konturlar → qizil ramka va ogohlantirish

Farq ikki joyda chiqadi: odam **turgan** va **kelgan** joyda — kadrlar orasidagi harakat izi.

Harakat detektor — to'liq kod

```
prev = None
while True:
    ok, frame = cap.read()
    g = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    g = cv2.GaussianBlur(g, (21, 21), 0)
    if prev is None:
        prev = g; continue
    diff = cv2.absdiff(prev, g)
    _, th = cv2.threshold(diff, 25, 255,
                          cv2.THRESH_BINARY)
    th = cv2.dilate(th, None, iterations=3)
    cnts, _ = cv2.findContours(th,
                               cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
    for c in cnts:
        if cv2.contourArea(c) > 800:
            x, y, w, h = cv2.boundingRect(c)
            cv2.rectangle(frame, (x, y),
                          (x + w, y + h), (0, 0, 255), 3)
    prev = g
```



25

Sezgirlik: kichik son — mayda harakatni ham sezadi



800

Minimal maydon: soya va shovqinni o'tkazib yuboradi

Yuzni aniqlash: Haar kaskad

```
path = cv2.data.harcascades + \
        "haarcascade_frontalface_default.xml"
face_det = cv2.CascadeClassifier(path)

while True:
    ok, frame = cap.read()
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_det.detectMultiScale(
        gray, scaleFactor=1.1, minNeighbors=5)
    for (x, y, w, h) in faces:
        cv2.rectangle(frame, (x, y),
            (x + w, y + h), (0, 255, 0), 2)
    cv2.putText(frame, f"Yuzlar: {len(faces)}",
        (20, 40), cv2.FONT_HERSHEY_SIMPLEX,
        1, (0, 255, 0), 2)
```

Tayyor model

OpenCV ichida keladi — yuklab olish shart emas

minNeighbors

Kattaroq — xato kamroq, lekin ba'zi yuzlar o'tib ketadi

Jonli demo

Natijani darsda kamerada ko'rsatamiz

05

Amaliyot va yakun

Tez-tez uchraydigan xatolar · topshiriqlar · xulosa 🕒 15 daqiqa

Tez-tez uchraydigan xatolar

Muammo	Sabab	Yechim
Kamera ochilmaydi	Boshqa dastur band qilgan yoki indeks xato	Zoom/Teams'ni yoping, VideoCapture(1)
Oyna qotib qoladi	waitKey() yo'q	Siklga cv2.waitKey(1) qo'shing
Ranglar g'alati	BGR va RGB aralashgan	cv2.COLOR_BGR2RGB bilan o'tkazing
frame — None	read() muvaffaqiyatsiz	ok ni tekshirib, break qiling
Video fayl bo'sh	O'lcham mos emas	Kadrni VideoWriter o'lchamiga resize
Juda sekin	Kadr juda katta	resize bilan kichraytiring

Mustaqil topshiriqlar

1 · OSON

Selfi-kamera

Ko'zgu effekti, ekranda ismingiz,
s — rasm saqlash

2 · O'RTA

Rang kuzatuvchi

Ko'k narsani toping va markaziga
nuqta chizing; u yurganda
iz qoldirsin

3 · QIYIN

Qo'riqchi kamera

Harakat bo'lsa, vaqt yozilgan
rasmni avtomatik saqlasin

Maslahat: markaz — `cv2.moments(c)` yordamida: $cx = m10 / m00$, $cy = m01 / m00$

Bugun nimalarni o'rgandik

`VideoCapture`

kamerani ochish

`read / imshow`

kadr olish va ko'rsatish

`cvtColor`

BGR → kulrang / HSV

`GaussianBlur` · `Canny`

filtr va chegaralar

`inRange`

rang bo'yicha maska

`findContours`

narsalarni ajratish

`absdiff`

harakatni topish

`CascadeClassifier`

yuzni aniqlash

8 ta funksiya — va kompyuteringiz endi **ko'radi**.

Savollar?

Rasmiy hujjatlar va qo'llanmalar: docs.opencv.org

[Ma'ruzachi ismi] · [Telegram / e-mail]